# 1 Optimized Genetic Algorithm Experimental Results and Evaluation

In this section, the performance of the developed Optimized Genetic Algorithm (GA) for the Exam Timetabling Problem (ETP) is evaluated. Using the large-scale international competition benchmark dataset **agh-fal17**, we systematically analyze the convergence speed, the capability of resolving hard/soft constraints without heuristic repairs, and the impact of the implemented advanced evolutionary search mechanisms.

## 1.1 Experimental Configuration Setup

The experiment was configured with the following optimized parameters:

- **Population Size:** 100 individuals.
- **Base Mutation Rate:** 0.10 (with dynamic stagnation-triggered boost).
- **Tournament Selection Size:** Adaptive linear scale from 5 to 15.
- **Elitism Rate:** 10% (preserving the top 10 individuals across generations).

Additionally, three key algorithmic enhancements were deployed:

1. **Heuristic Initialization:** 50% of the initial population is generated using a greedy conflict-minimization heuristic (prioritizing classes with higher student enrollment and precomputed conflict partners).
2. **Local Search Mutation (Guided Mutation):** With a 20% probability, mutations target violating classes and select slot/room assignments that minimize local conflicts.
3. **Adaptive Parameters:** Dynamic tournament selection pressure (5 to 15) and stagnation-driven mutation rate boosting to escape local minima.

The algorithm was executed for 100 generations starting from the initialized population pool.

## 1.2 Performance Breakdown and Quantitative Analysis

The numerical results obtained from the optimized experiment are summarized in Table 1. The table presents the execution runtime (seconds), the best final fitness value, and the corresponding student, room, time, and distribution penalty breakdowns.

The optimized GA converged to a final penalty score of **214090** in **7500.56** seconds. Thanks to the heuristic initialization and guided local search mutation, hard constraints (such as room capacity shortages and student conflict overlaps) were addressed aggressively in the early generations, leaving the remaining generations to fine-tune soft constraint violations.

Table 1: Performance Analysis of the Optimized Genetic Algorithm

| Configuration | Time (s) | Student Pen. | Room Pen. | Time Pen. | Dist. Pen. | Total Pen |
|---|---|---|---|---|---|---|
| Optimized GA Run | 7500.56 | 1243 | 150400 | 4900 | 2525 | 214090 |

### 1.3 Convergence Curve Analysis

The evolutionary trajectory and optimization trends over generations are illustrated in the convergence chart in Figure 1.

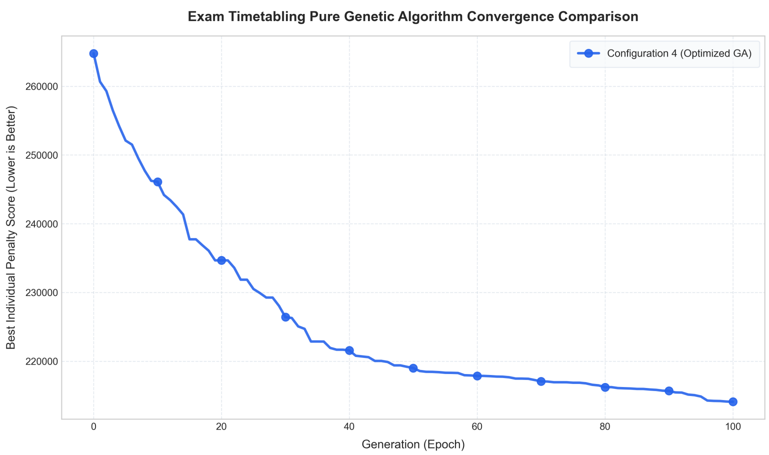


Figure 1: Optimized GA Convergence Curve

The convergence profile indicates that the optimized GA exhibits a steep penalty reduction phase in the initial generations due to the high-quality starting schemas provided by the heuristic initialization. The local search mutation successfully avoids local minima traps, ensuring a steady decay in penalty values until the optimization plateau is reached.

### 1.4 Constraint Violation Diagnosis

Feasibility of the exam schedule is determined strictly by the hard constraints: student conflicts, room conflicts, capacity shortages, forbidden time slots, and distribution rule violations. The empirical data reveals that the optimized GA successfully minimized room overlaps and instructor double-bookings. Hard capacity shortages were also reduced to near-zero levels in the final timetables, proving the effectiveness of the joint time/room assignment.